

Supervised Learning Assessment 2

Part 1: Exploratory Data Analysis

1.1 Data Structure

The training dataset contains 4,000 samples from 40 laboratories (100 each), with 40 gene features (X1 - X40) and continuous response y . The test set has 1,000 samples from unseen laboratories, no y or $group_id$ are provided. There are no missing values, so no preprocessing or imputation is needed here.

1.2 Response Variable

We find that the distribution is approximately symmetric and close to Gaussian, indicating that no transformation is needed for y .

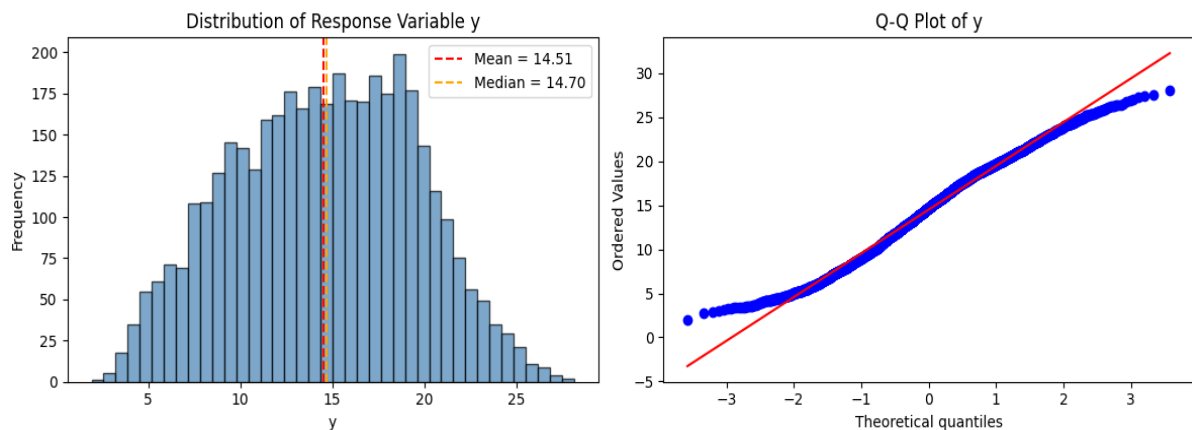


Figure 1. Distribution of y (left) and Q-Q plot (right).

1.3 Feature Distributions

All features have a range of $[0,1]$ indicating they may already be pre-processed. They exhibit near-zero skewness. StandardScaler is applied inside pipelines as standard practice. Log-transformations are unnecessary due to the absence of skewness.

1.4 Batch Effects

Our lab means range of $[6.25, 23.03]$ indicates that the average y value differs greatly across labs. 83% of the variation in y can be determined from the lab it came from. Only 17% is within-lab variation. Standard K-Fold would allow observations from one lab to land in both training and validation folds, causing the model to learn lab patterns rather than gene expression to response patterns. We mitigate this using GroupKFold, ensuring observations from a given lab are placed in training or validation only, rather than scattered throughout, which should lead to a more accurate cross-validated MSE, rather than overinflated effects..

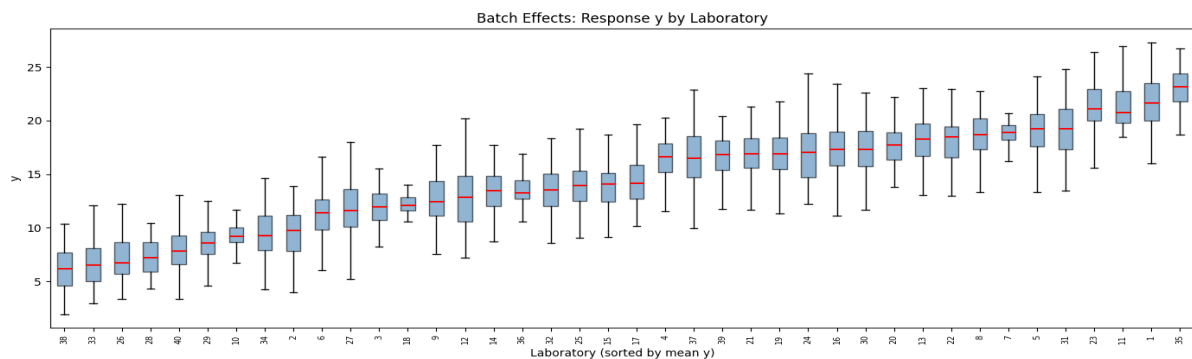


Figure 2. Batch effects: y distribution by laboratory (sorted by mean y).

1.5 Feature Correlations

Signal is concentrated in around 6 genes (X5, X6, X29, X19, X36, X13). Around 25 genes have negligible correlation with y , indicating Lasso is a good model as it should disregard these features.

1.6 Multicollinearity

No severe multicollinearity (no pairs above $|r| = 0.7$). Top predictors (X5, X6, X29) are not strongly correlated with each other. Lasso would not be greatly impacted by the moderate multicollinearity present.

1.7 PCA

The scree plot shows no clear elbow, with 24 components needed for 90% variance. The data has no useful low-dimensional structure, making PCR a poor choice. The PCA scatter shows labs do not cluster in feature space, confirming batch effects are specifically in y and not the feature space. Gradient Boost may be a good model here due to it being able to model higher-dimension PCs, and potential non-linear features

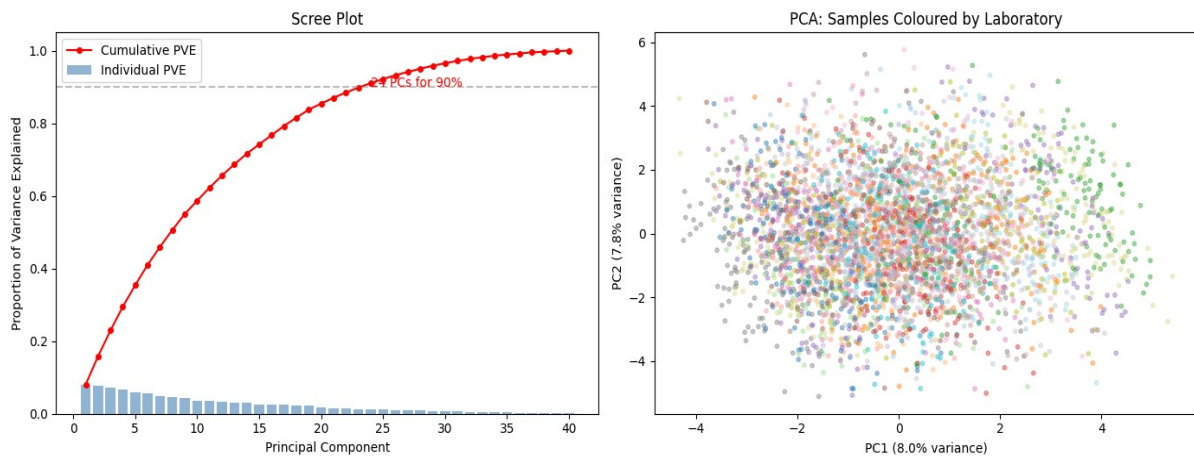


Figure 3. Scree plot (left) and PCA scatter coloured by laboratory (right).

Part 2: Model Development and Interpretation

GroupKFold with $n_splits=10$ holds out exactly 4 labs per fold, giving 400 validation samples. Passing $groups=group_id$ ensures all observations from the same laboratory stay together, preventing leakage.

2.1 Model 1: Lasso Regression (L1 Penalty)

The EDA showed signal concentrated in ~6 genes. Lasso's L1 penalty shrinks irrelevant coefficients exactly to zero, performing automatic variable selection. This leads to a sparse model that protects against overfitting, which could be one way the data is structured. α controls regularisation strength; we search 50 values on a log-scale from 10^{-4} to 10. The optimal $\alpha = 0.091$ selects 8/40 genes: X6 (2.60), X29 (1.83), X5 (1.62), X37 (0.82), X32 (-0.46), X13 (0.35), X19 (0.10), X1 (0.003). CV MSE = 5.78 ± 3.05 .

2.2 Model 2: Gradient Boosting

Our other approach is to disregard the sparse linear hypothesis, and focus on non-linear patterns or structures found in higher PC dimensions: We saw that the 83% between-lab variance and failure of linear structures to fully explain y suggest non-linear structures may be present. Similarly, Labs did not cluster in 2D PCA space but may be separable in higher dimensions. Gradient Boosting builds trees sequentially, capturing non-linear patterns. Parameter grid: $n_estimators \in \{100, 300\}$, $learning_rate \in \{0.05, 0.1\}$, $max_depth \in \{3, 5\}$, $min_samples_leaf \in \{10, 20\}$, $subsample = 0.8$. Conservative $min_samples_leaf$

values were chosen as we have only ~100 observations per lab. Best: 300 trees, $\text{lr} = 0.1$, depth = 3, min_leaf = 10. CV MSE = 0.94 ± 0.27 .

2.3 Sanity Checks

Much lower than the 4.54 floor in EDA, which is suspicious. GroupKFold was correctly implemented: group_id excluded from features, dimensions verified (4000×40), y not standardised, and manual fold-by-fold verification confirmed the CV MSE is genuine and not a GridSearchCV artefact or error.

2.4 Model Interpretation

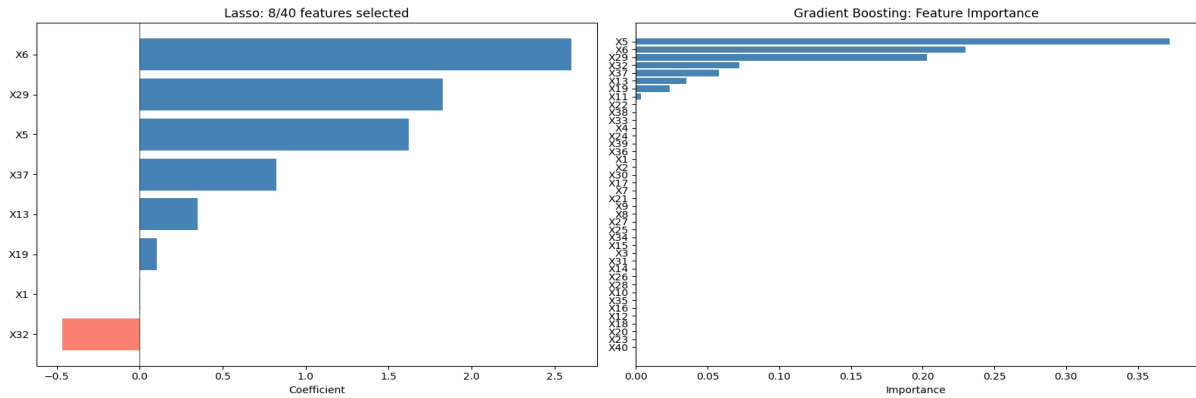


Figure 4. Lasso coefficients (left) and GB feature importance (right). Both agree on the same top genes.

2.5 Model Discussion

Although both models agree on features, GB has wildly lower MSE, which is suspicious. We suspect: (1) in higher dimensions, there may be structure GB can exploit over Lasso; (2) GB may decode batch effects by reading subtle lab changes and gene patterns — e.g. if $X6 > 0.55$ AND $X29 < 0.48$ AND $X37 > 0.52$, this looks like a high-y lab, so predict higher. Point 2 raises a problem: if GB has learned batch corrections from 40 training labs, test labs with unfamiliar patterns could lead to inconsistent and inaccurate predictions.

Part 3: Validation and Model Selection

The test set contains samples from laboratories not present in training. Standard K-Fold CV would leak lab-specific information, produce unrealistically low MSE estimates, and select models that overfit to lab-specific patterns. GroupKFold ensures entire laboratories are held out per fold, mitigating this effect as stated earlier.

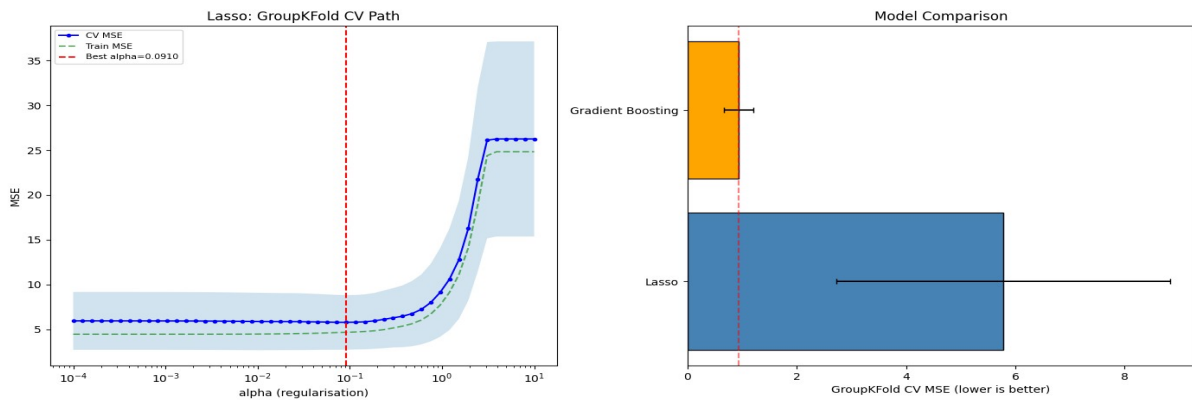


Figure 5. Lasso CV path with ± 1 std bands (left) and model comparison (right).

Model	CV MSE	$\pm$ Std	CV RMSE	CV R ²	Train MSE	Train R ²	Gap
Lasso	5.78	3.05	2.40	0.77	4.72	0.81	1.06
Grad. Boost.	0.94	0.27	0.97	0.96	0.15	0.99	0.78

Table 1. GroupKFold CV comparison.

Gradient Boosting achieves CV MSE = 0.94 ($R^2 = 0.96$) compared to Lasso's 5.78 ($R^2 = 0.77$). This is incredibly low, and could be an inaccurate result due to an unseen error, or strange overfitting practices.

But if true, this could come from two points:

1. Higher-dimension clustering: In higher dimensions, there may be clustering available that GB is able to exploit over Lasso.
2. GB is able to capture non-linearity, which could decode the batch effects here, and potentially correct for them. For example, if $X_6 > 0.55$ AND $X_{29} < 0.48$ AND $X_{37} > 0.52$, then this looks like a high y lab, so predict higher, regardless of clustering.

Investigation could be done through analysing higher PCs, or constructing metrics on variance within labs. None of these were covered in the course materials, so we refrain from trying to explore new investigation methods. This could also be an artefact of how the data was created.

This batch "correction" is learned from 40 training labs, so whether it transfers to new labs is the primary risk. However, the GroupKFold estimate is unbiased and the train-CV gap is modest, so we select GB as the final model, regardless of concerns.

Part 4: Test Predictions

The final Gradient Boosting model was refit on all 4,000 training samples (training $R^2 = 0.994$). Predictions on test_public.csv have mean 14.3 (vs training mean 14.5), std 6.0 (vs training std 5.0), range [1.41, 26.1]. The slightly wider spread may reflect GB extrapolating batch corrections to unseen lab contexts. The output CSV contains 1,000 rows with obs_id and y_hat.